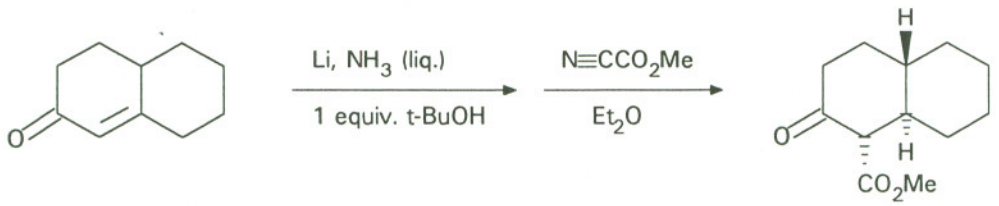
ORGANIC CUMULATIVE EXAMINATION
October 3, 1992

Propose a reasonable step-by-step mechanism for three of the following reactions. Show the structural formula for each important mechanistic intermediate. Describe how your mechanism explains the stereochemical result of the reaction if that is pertinent. (Reagents and solvents shown over and under the same arrow are simultaneously present. Reagents shown over and under a second arrow are added later, but without isolation of a product from the reaction brought about by the first arrow. All reactions were worked up in the usual manner.) **Show clearly which three reactions you want graded. If you do not, reactions 1, 2, and 3 will be graded.**

1.



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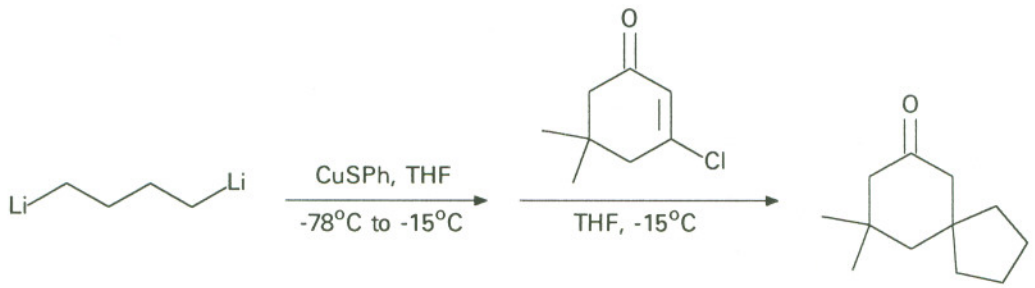

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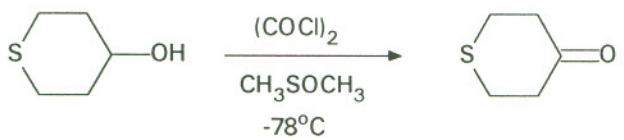

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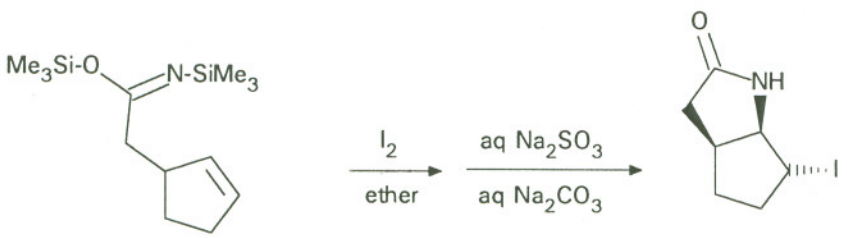

[Answers: Organic Syntheses, Volume 70, 1991, pp 256, 204, 120, 101 and references therein.]